EXPERIMENT 18: Statistical Analysis of Age and Body Fat Using Pandas

1. AIM

To calculate the mean, median, and standard deviation of age and body fat percentage using Pandas, and draw boxplots, a scatter plot, and a Q-Q plot.

2. ALGORITHM

1. **Import Pandas, Matplotlib, and SciPy libraries.**
2. **Store the age and body fat data.**
3. **Calculate mean, median, and standard deviation.**
4. **Draw boxplots for age and body fat.**
5. **Draw a scatter plot and Q-Q plots.**
6. **Display the results.**

3. CODE

Experiment 18 Python Code

```
import pandas as pd
import matplotlib.pyplot as plt
from scipy import stats

age = [23, 23, 27, 27, 39, 41, 47, 49, 50,
    52, 54, 54, 56, 57, 58, 58, 60, 61]

fat = [9.5, 26.5, 7.8, 17.8, 31.4, 25.9, 27.4, 27.2, 31.2,
    34.6, 42.5, 28.8, 33.4, 30.2, 34.1, 32.9, 41.2, 35.7]

df = pd.DataFrame({"Age": age, "%Fat": fat})

print(df.agg(["mean", "median", "std"]))

df.boxplot()
```

```
plt.title("Boxplots of Age and Body Fat")

plt.show()


plt.scatter(age, fat)

plt.xlabel("Age")

plt.ylabel("Body Fat %")

plt.title("Age vs Body Fat")

plt.show()


plt.subplot(1, 2, 1)

stats.probplot(age, dist="norm", plot=plt)

plt.title("Q-Q Plot of Age")


plt.subplot(1, 2, 2)

stats.probplot(fat, dist="norm", plot=plt)

plt.title("Q-Q Plot of Body Fat")


plt.tight_layout()

plt.show()
```

4. INPUT

The program uses the 18 age and body fat observations given in the question.

5. OUTPUT

```
>
=========== RESTART: C:/Users/arunp/OneDrive/Documents/FDS/EXP 18.py ===========
Mean:
Age          46.444444
Body Fat     28.783333
dtype: float64

Median:
Age          51.0
Body Fat     30.7
dtype: float64

Standard Deviation:
Age          13.218624
Body Fat      9.254395
dtype: float64
```



6. RESULT

Thus, the statistical measures were calculated, and the boxplots, scatter plot, and Q-Q plots were generated successfully.